DSP Report (1)

Experiment (1): Digital Waveforms Generation

Using MATLAB software Generate and plot the following signals in time domain:

i. Sine wave with frequency = 1 KHz and amplitude = 5 V
ii. Sawtooth wave with frequency = 2 KHz and amplitude = 6 V
iii. Triangle wave with frequency = 3 KHz and amplitude = 7 V
iv. Square wave with frequency = 4 KHz, amplitude = 8 V and duty cycle = 25%
v. Impulse function time delayed by 1 ms
vi. Unit step function time delayed by 2 ms
vii. Ramp function time advanced by 3 ms
viii. Rectangular pulse with width = 4 ms and time advanced by 4 ms
ix. Triangle pulse with width = 4 ms and time delayed by 5 ms
x. Sinc function where T = 0.5 ms